

JEE MAIN REVISION MASTER: LIMITS, CONTINUITY AND DIFFERENTIABILITY 2026 3 QUESTIONS 5% WEIGHTAGE ↓25.04%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 29 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Continuity & Differentiability at a Point	19 PYQs	Continuity & Differentiability
High	Evaluation of Limits & L'Hopital's Rule	13 PYQs	Limits

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Continuity & Differentiability at a Point	$\text{Continuous at } a \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a), \quad \text{Differentiability} \implies \text{Continuity}$
Evaluation of Limits & L'Hopital's Rule	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1, \quad \lim_{x \rightarrow a} f(x)^{g(x)} = e^{\lim_{x \rightarrow a} (f(x) - 1)g(x)} \text{ (for } 1^\infty \text{)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Continuity & Differentiability at a Point (Very High Yield — 19 PYQs)

Core Principle: Epsilon-delta and limit criteria for continuity, left/right hand derivatives, and graphical identification of non-differentiable sharp points.

■■ **Caution / JEE Pitfall:** Differentiability implies continuity, but continuity DOES NOT imply differentiability (e.g. $f(x) = |x|$ is continuous at $x=0$ but not differentiable).

■ **Speed Trick:** The greatest integer function $[x]$ is discontinuous at all integer points $x \in \mathbb{Z}$.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): "If $f(x) = \begin{cases} ax+7 & \text{if } x < 1 \\ 3x-1 & \text{if } x = 1 \\ b/(x+3) & \text{if } x > 1 \end{cases}$ is continuous at $x=1$." [Ans: D]
- [2025 Shift] Q#1 (MCQ): "Match the following: In the following, $[x]$ denotes the greatest integer less than or equal to x .
Column - I Column - II (a) $x | x | x | x \dots$ " [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The function $f(x) = e^x + ax, x < 0$ and $b(x-1)^2, x \geq 0$ is differentiable at $x=0$. Then..." [Ans: N/A]

3.2 Evaluation of Limits & L'Hopital's Rule (High Yield — 13 PYQs)

Core Principle: Resolution of indeterminate forms ($0/0$, ∞/∞ , 1^∞), standard trigonometric and exponential limits, and Series expansion.

■ **Caution / JEE Pitfall:** Before applying L'Hopital's rule, ALWAYS check that the limit is in an indeterminate form ($0/0$ or ∞/∞). Differentiating a determinate form produces an incorrect answer.

■ **Speed Trick:** Maclaurin expansions are often faster than repeated L'Hopital: $\sin x = x - (x^3/6)$, $\cos x = 1 - (x^2/2)$, $e^x = 1 + x + (x^2/2)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If $f(x) = x^2 - 1$ if $x \geq 2$ and $x + 1$ if $x < 2$, then $\lim_{x \rightarrow 2} f(x) =$..." [Ans: B]
- [2026 Shift] Q#2 (MCQ): "If $\lim_{x \rightarrow 2} (\frac{x^2 - ax - 3bx - 3}{x - 2}) = 5$, then $a + b =$..." [Ans: C]
- [2026 Shift] Q#3 (MCQ): "If $f(x) = \begin{cases} ax + 7 & \text{if } x < 1 \\ 3x - 1 & \text{if } x = 1 \\ b/x + 3 & \text{if } x > 1 \end{cases}$ is continuous at $x = 1$, then $a + b =$..." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If $f(x) = x^2 - 1$ if $x \geq 2$ and $x + 1$ if $x < 2$, then $\lim_{x \rightarrow 2} f(x) =$

- (A) 3
- (B) 5 [Correct]
- (C) 7
- (D) 9

Correct Answer: B

Solution Analysis: For $x \rightarrow 1$, the values of x are near 1, which is less than 2. So we use $f(x) = x + 1$. Hence, $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2$. Now for $x \rightarrow 2$, we check both sides. For $x < 2$, $f(x) = x + 1$. So, $\lim_{x \rightarrow 2^-} f(x) = 2 + 1 = 3$. For $x \geq 2$, $f(x) = x^2 - 1$. So, $\lim_{x \rightarrow 2^+} f(x) = 2^2 - 1 = 4 - 1 = 3$. Since both left-hand and right-hand limits are equal, $\lim_{x \rightarrow 2} f(x) = 3$. Now, $\lim_{x \rightarrow 2} f(x) = 3$. So the correct

Benchmark Problem 2: KCET 2025 (Q#1)

Match the following: In the following, $[x]$ denotes the greatest integer less than or equal to x . Column - I Column - II (a) $x | x |$ (i) continuous in $(-1, 1)$ (b) $|x| |x| \sqrt{|x|}$ (ii) differentiable in $(-1, 1)$ (c) $x + [x]$ (iii) strictly increasing in $(-1, 1)$ (d) $|x - 1| + |x + 1|$ (iv) not differentiable at, at least one point in $(-1, 1)$

- (A) a-i, b-ii, c-iv, d-iii
- (B) a-iv, b-iii, c-i, d-ii
- (C) a - ii, b - iv, c - iii, d - i
- (D) a - iii, b - ii, c - iv, d - i

Correct Answer: None

Solution Analysis: (a) $x|x|$ $f(x) = x^2$, $x \geq 0$ and $-x^2$, $x < 0$. Differentiable in $(-1, 1)$. (b) $\sqrt{|x|} = \sqrt{x}$, $x \geq 0$ and $\sqrt{-x}$, $x < 0$. Not differentiable at $x = 0$. (c) $x + [x]$ strictly increasing in $(-1, 1)$. (d) $|x - 1| + |x + 1| = \begin{cases} -x + 1 + x + 1 = 2 & \text{if } x < -1 \\ -x + 1 + x - 1 = 0 & \text{if } -1 \leq x \leq 1 \\ -x + 1 + x + 1 = 2 & \text{if } x > 1 \end{cases}$. Continuous in $(-1, 1)$. $\therefore$ (A) a-i, b-ii, c-iv, d-iii